

QuaRM: Accounting for Data-Quality Debt Across Interacting Defects and Analytical Workloads

Supplementary Material and Artifact Guide

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This supplement expands the experimental record for the ICDE 2027 submission. It is not an appendix to the 12-page paper and contains no claim required to understand the main argument. Every table is generated directly from the checked-in result CSVs by `scripts/make_icde_paper_assets.py`.

I. ARTIFACT MAP

The public release must expose the following structure:

- `src/quarm`: exact attribution, sampling, ML, and relational backends;
- `experiments`: frozen study drivers and paired-design ablation;
- `artifacts/icde_results`: raw coalition observations, allocations, interactions, query allocations, scale, sampling, and SHA-256 manifests;
- `artifacts/results`: ML confirmation evidence and manifest;
- `paper/icde2027`: IEEEtran sources and generated evidence tables;
- `tests`: deterministic unit and integration tests; and
- `scripts`: official-data fetcher, asset generator, PDF QA, and submission audit.

Raw NYC inputs are not duplicated in the repository. `scripts/fetch_public_data.py` downloads them from the official TLC endpoint and rejects files whose hashes differ from the frozen January 2025 snapshot.

II. MECHANISMS AND QUERY WORKLOAD

The eight workload templates are: average order value by customer segment; customer count by region; daily revenue; order count by status; refund rate by region; revenue by product category; monthly revenue; and revenue by region. Each answer is aligned on group keys with absent groups filled by zero. The loss is clipped normalized L_1 error, and workload loss is the unweighted mean. Query-level allocations are retained in `query_attributions.csv` and `nyc_query_attributions.csv`.

III. COMPLETE RELATIONAL RESULTS

Exact Shapley efficiency residual is zero for every relational population. The normalized Banzhaf residuals are -1.89 risk points for NYC and between -1.40 and -1.56 for the generated references. The largest absolute pairwise interactions are 5.62 (NYC), 5.87 (Uniform), 5.69 (Skewed), and 5.90 (Seasonal)

points. In all four, missing measures and duplicate facts form the largest-magnitude pair.

IV. SCALE AND ESTIMATOR DETAILS

The exact row-scale experiment performs 64 coalition materializations and 512 SQL query executions per size. Wall times are 9.55 s at 10K facts, 22.17 s at 100K, and 97.22 s at 1M. These values include Python intervention, DuckDB ingestion, query evaluation, and answer comparison. They were collected on the environment recorded in `artifacts/icde_results/manifest.json`.

For permutation sampling, each random order visits $m + 1$ prefixes. A process-local cache ensures a coalition is evaluated at most once. The result object records sample count, marginal matrix, standard errors, exact coalition count, unique coalition count, and the simultaneous Hoeffding radius. The controlled surface contains unequal main effects, eight pair interactions, and three triple interactions followed by a logistic map; all 4,096 losses are therefore in $(0, 1)$ and available for exact ground truth.

V. ML CONFIRMATION

The ML study follows the same paired-coalition contract with four mechanisms: missing cells, numeric feature noise, target noise, and duplicate rows. Every setting evaluates all 16 coalitions for 12 paired train/test splits. Seven of eight aggregate debts are positive; the nonlinear-synthetic/logistic setting is negative because feature noise acts as regularization for the fixed misspecified learner. This signed case is retained rather than clipped.

VI. REPRODUCTION PROTOCOL

On Windows PowerShell, after creating the environment and installing the project:

```
.venv\Scripts\python.exe -m pytest
.venv\Scripts\python.exe scripts\fetch_public_data.py
.venv\Scripts\python.exe experiments\run_study.py
.venv\Scripts\python.exe experiments\run_icde_study.py
.venv\Scripts\python.exe experiments\run_icde_real.py
.venv\Scripts\python.exe experiments\run_sampling_scalability.py
.venv\Scripts\python.exe experiments\analyze_paired_design.py
.venv\Scripts\python.exe scripts\make_icde_paper_assets.py
```

The expensive full relational study requires approximately 15 minutes on the recorded laptop-class environment; the exact NYC study requires approximately four minutes. The public release should preserve the evidence CSVs so reviewers can

TABLE I

RELATIONAL EXPERIMENTAL CONTRACT. ALL SELECTED UNITS ARE DETERMINISTIC FUNCTIONS OF THE MASTER SEED, REPEAT, AND MECHANISM NAME.

Mechanism	unit selected at severity θ	transformation
missing measures	monetary fact cells	replace with SQL NULL
value noise	monetary fact cells	multiply by a stochastic scale perturbation
orphan foreign keys	fact foreign-key cells	replace with an absent identifier
stale timestamps	fact event times	shift backward according to the fixed operator
dimension drift	selected dimension members	replace the analytical category/region label
duplicate facts	fact rows	append exact row copies after value-level mechanisms

TABLE II

COMPLETE RELATIONAL SHAPLEY ALLOCATIONS. VALUES AND INTERVALS ARE WORKLOAD RISK POINTS; RANK IS THE PAIRED-BOOTSTRAP PROBABILITY OF BEING THE LARGEST ACCOUNT.

Reference	mechanism	debt	95% low	95% high	rank-one (%)
NYC Taxi	dimension drift	9.56	8.45	10.73	100.0
NYC Taxi	stale time	3.32	3.30	3.35	0.0
NYC Taxi	duplicate facts	2.55	2.00	3.12	0.0
NYC Taxi	orphan keys	1.63	0.95	2.51	0.0
NYC Taxi	value noise	1.22	1.00	1.59	0.0
NYC Taxi	missing measures	0.91	0.86	0.96	0.0
Uniform	duplicate facts	2.27	2.22	2.33	100.0
Uniform	value noise	1.94	1.83	2.05	0.0
Uniform	stale time	1.87	1.84	1.90	0.0
Uniform	orphan keys	0.43	0.40	0.46	0.0
Uniform	dimension drift	0.41	0.33	0.51	0.0
Uniform	missing measures	0.27	0.21	0.32	0.0
Skewed	duplicate facts	2.39	2.32	2.46	94.7
Skewed	value noise	1.98	1.89	2.06	0.0
Skewed	stale time	1.89	1.87	1.92	0.0
Skewed	dimension drift	1.77	1.06	2.56	5.3
Skewed	orphan keys	0.57	0.51	0.63	0.0
Skewed	missing measures	0.24	0.18	0.30	0.0
Seasonal	duplicate facts	2.26	2.20	2.32	100.0
Seasonal	stale time	1.91	1.87	1.94	0.0
Seasonal	value noise	1.90	1.82	1.97	0.0
Seasonal	dimension drift	0.44	0.39	0.50	0.0
Seasonal	orphan keys	0.44	0.42	0.46	0.0
Seasonal	missing measures	0.28	0.23	0.34	0.0

regenerate all manuscript tables and figures without rerunning expensive workloads, then selectively reproduce any study.

VII. ARTIFACT CLAIMS AND CHECKS

The artifact supports the following independently checkable claims:

- 1) all exact games contain every coalition exactly once per repeat;
- 2) Shapley allocations close aggregate debt to floating-point precision;
- 3) identical master seeds replay identical corruption selections and losses;
- 4) a mechanism’s random realization is invariant to coalition membership;
- 5) the NYC inputs match pinned hashes from the official source;

- 6) every paper number is generated from machine-readable evidence;
- 7) every final PDF page is rendered and checked for page-bound violations; and
- 8) the wheel, source distribution, tests, and paper sources are included in the release manifest.

The release should be assigned a DOI or immutable repository tag before submission. A mutable branch URL is insufficient for archival review.

ACKNOWLEDGMENT OF AI-GENERATED CONTENT

OpenAI Codex was used to generate and revise portions of the Python implementation, experimental orchestration, figures, supplementary material, and manuscript text. Before submission, the listed authors must independently inspect the source, reproduce the evidence, verify the claims, and accept

TABLE III
COMPLETE ML CONFIRMATION STUDY AT 15% PREVALENCE. DEBT AND MAXIMUM INTERACTION ARE BOUNDED CLEAN-TEST RISK POINTS.

Dataset	model	rows	debt	top account	top prob. (%)	max $ I $	full-context best
breast cancer	logistic	569	3.80	target noise	98.2	1.42	target noise
breast cancer	forest	569	1.12	missing cells	68.3	0.39	missing cells
wine	logistic	178	6.48	target noise	99.8	2.03	target noise
wine	forest	178	1.70	target noise	78.2	0.54	target noise
diabetes	ridge	442	4.21	feature noise	100.0	0.96	feature noise
diabetes	forest	442	1.42	feature noise	59.2	0.31	feature noise
synthetic nonlinear	logistic	900	-1.33	missing cells	54.8	0.58	target noise
synthetic nonlinear	forest	900	7.28	feature noise	99.1	1.65	feature noise

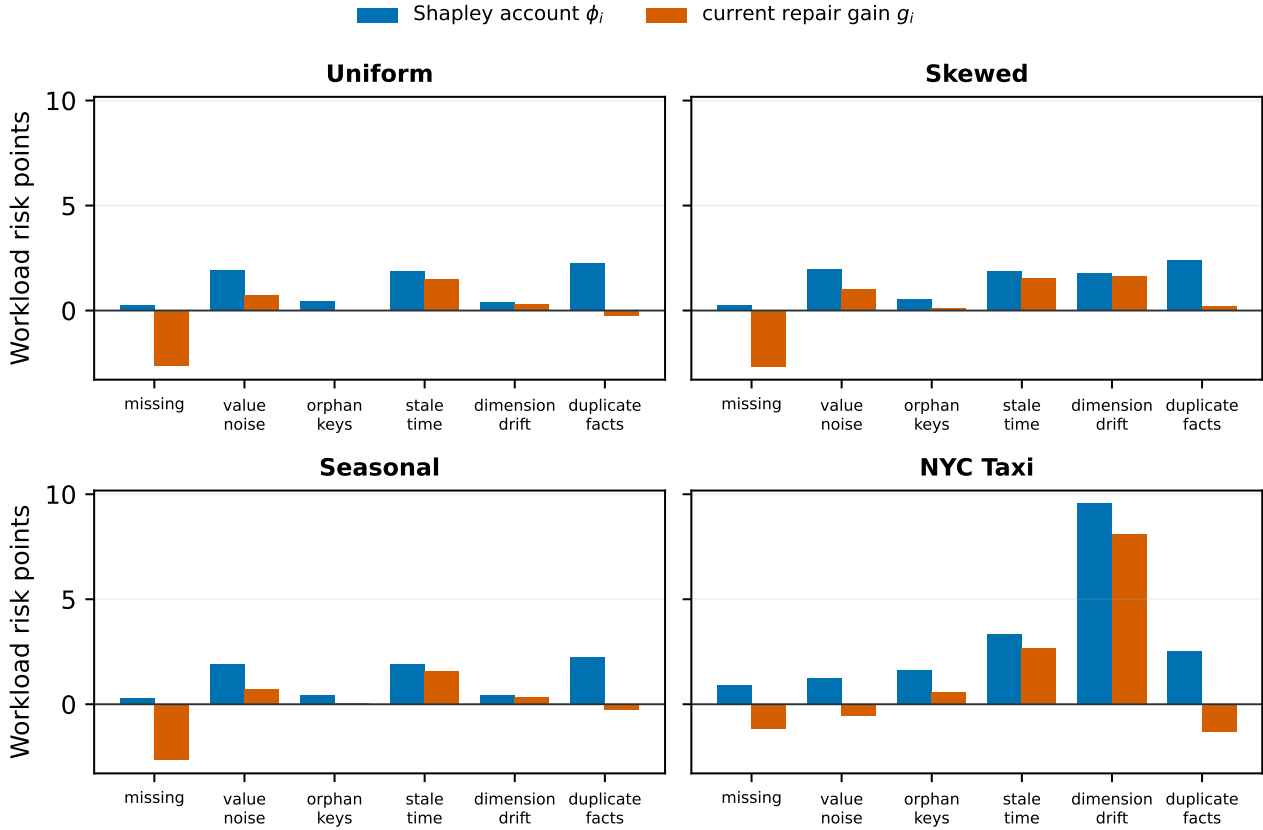


Fig. 1. The complete accounting/intervention comparison. The underlying coalition means and every full-context marginal are available in the raw observations.

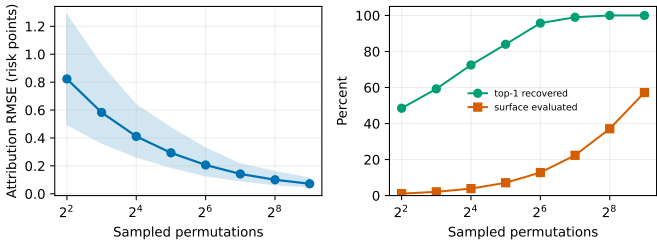


Fig. 2. Estimator convergence and actual coalition coverage for the 12-mechanism study.

responsibility for the complete work. No AI system is an author.

TABLE IV
CONTROLLED 12-MECHANISM PERMUTATION STUDY (400 TRIALS PER BUDGET).

B	median RMSE	top-one (%)	surface (%)
4	0.823	48.5	1.1
8	0.583	59.2	2.1
16	0.411	72.5	3.9
32	0.294	84.0	7.1
64	0.206	95.8	12.8
128	0.142	99.0	22.4
256	0.100	100.0	37.1
512	0.072	100.0	57.2